

Unit 405 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 42.6 ac. 20 plots, BAF 20, variable-radius cruise on a 300 ft grid, 236 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	236 sq ft/ac (1 SE of mean, BA 2.3 %)	Type	Sierran mixed conifer
Trees per acre	61	CWHR size / density	5 / D (cover 95 %)
QMD	26.6 in	SDI	278 (47 % of max 593)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	235 sq ft/ac 27.0 in	BA QMD	1 sq ft/ac 9.5 in
TPA	59	TPA	2
Volume	47.1 MBF/ac (9389 cu ft)	Volume	0.4 green tons/ac (16 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 176 sq ft/ac BA	Vol. sampling error	12.6 % at 95 % (CV 27 %, t = 2.09, n = 20)
Species (BA)	WF 75, PP 57, IC 38, DF 37, SP 29	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 4
		Sale volume share	3.7 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.7 % at 95 % (CV 10 %); reference, not the standard
		Measured trees	PP 57, WF 75, DF 37, SP 29, IC 38; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	20; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	3	5.3	3.0	59	0.3
12-15.9	10	9.4	10.0	234	1.2
16-19.9	20	11.0	20.0	558	2.8
20-23.9	22	8.6	22.0	710	3.6
24-27.9	34	9.3	34.0	1211	6.1
28-31.9	16	3.3	16.0	560	2.8
32-35.9	34	5.4	34.0	1320	6.6
36-39.9	23	3.0	23.0	1039	5.2
40-43.9	22	2.3	22.0	1059	5.3
44-47.9	14	1.2	14.0	689	3.5
48-51.9	9	0.7	9.0	426	2.1
52-55.9	12	0.8	12.0	583	2.9
56-59.9	6	0.3	6.0	355	1.8
60-63.9	11	0.6	11.0	604	3.0
All	236	61.0	236.0	9406	47.2

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	75	17.0	75.0	28.4	3005	15.1
PP	57	15.4	57.0	26.0	2603	13.1
IC	38	11.6	38.0	24.5	1262	6.3
DF	37	10.1	37.0	25.9	1326	6.7
SP	29	6.9	29.0	27.7	1210	6.1
All	236	61.0	236.0	26.6	9406	47.2

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 405 plots, 300 ft grid (orange: QA flag)

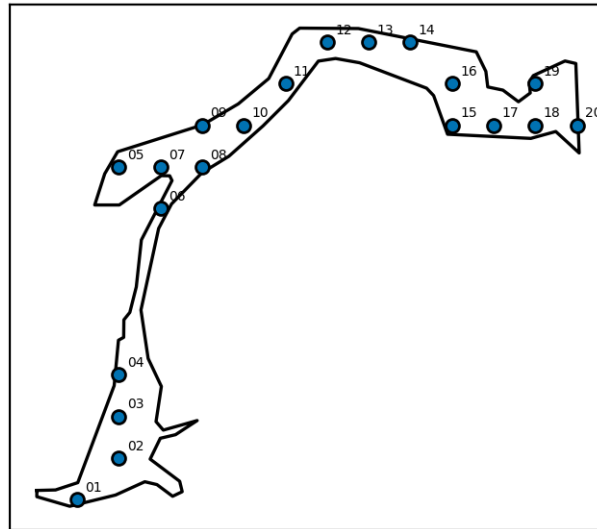


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
405-01	12	240	55	ok
405-02	12	240	70	ok
405-03	10	200	66	ok
405-04	12	240	59	ok
405-05	11	220	61	ok
405-06	11	220	48	ok
405-07	12	240	58	ok
405-08	12	240	48	ok
405-09	11	220	36	ok
405-10	12	240	65	ok
405-11	12	240	60	ok
405-12	13	260	60	ok
405-13	12	240	56	ok
405-14	15	300	37	ok
405-15	10	200	44	ok
405-16	14	280	49	ok
405-17	11	220	49	ok
405-18	11	220	53	ok
405-19	12	240	37	ok
405-20	11	220	50	ok